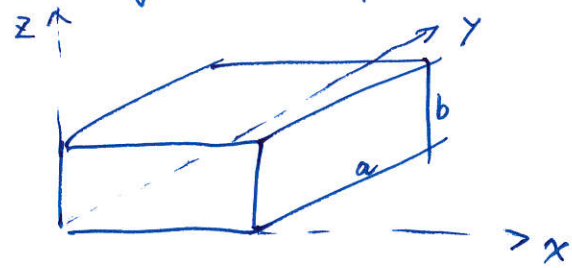
Discussion. March 29th 2016

Waveguide: Perfect Conductors



- Consider a hollow pipe infinite along x. The cross section is rectangular with sides a and b.

- We set the guide walls to be perfect conductors such that there will be no E or B field in the walls, and we don't need to worry about solving Maxwell equations there.

- The translational symmetry in x-direction motivates the ansatz $\vec{E}, \vec{B} \propto e^{ik_x x - i\omega t}$.

Since both outside and inside the pipe we have vacuum, it's just about solving vacuum Maxwell equations with special boundary conditions.

$$\begin{cases} \vec{\nabla} \times \vec{E} = i\omega \vec{B} \\ \vec{\nabla} \times \vec{B} = -\frac{i\omega}{c^2}\vec{E} \end{cases} \quad \text{w/} \quad \vec{E}_{\parallel} = 0$$

(The tangential components on the boundary vanish because ① $\vec{j} = \sigma \vec{E}$, perfect conductor $\sigma \to \infty$, to obtain finite j we need $E \to 0$; ② $\vec{E} = 0$ in the walls and tangential component must be continuous.)

*($\vec{E}_{\parallel} = 0$, example: on the $z=b$ surface, $\vec{E} = (0, 0, E_z)$; on the $y=0$ surface, $\vec{E} = (0, E_y, 0)$.)

- ~~Inside~~ One can either have 1) transverse magnetic wave satisfying $\vec{B} \perp \hat{x}$ or 2) transverse electric wave satisfying $\vec{E} \perp \hat{x}$. We focus on the first case.

- Inside the wave guide, $\vec{B} = (0, B_y, B_z)$, $\vec{E} = (E_x, E_y, E_z)$.
Writting out our maxwell equation is in components, we have

$$\begin{cases} -\frac{\partial}{\partial x} E_z + \frac{\partial}{\partial z} E_x = i\omega B_y \\ \text{(The } B_x \text{ equation is just zero.)} \\ \frac{\partial}{\partial x} E_y - \frac{\partial}{\partial y} E_x = i\omega B_z \end{cases} \Rightarrow \begin{cases} -ik_x E_z + \frac{\partial E_x}{\partial z} = i\omega B_y \\ ik_x E_y - \frac{\partial E_x}{\partial y} = i\omega B_z \end{cases}$$

and
$$\begin{cases} \dfrac{\partial B_z}{\partial y} - \dfrac{\partial B_y}{\partial z} = -\dfrac{i\omega}{c^2} E_x \\ -\dfrac{\partial B_z}{\partial x} + \dfrac{\partial B_x}{\partial z} = -\dfrac{i\omega}{c^2} E_y \\ \dfrac{\partial B_y}{\partial x} - \dfrac{\partial B_x}{\partial y} = -\dfrac{i\omega}{c^2} E_z \end{cases} \Rightarrow \begin{cases} \dfrac{\partial B_z}{\partial y} - \dfrac{\partial B_y}{\partial z} = -\dfrac{i\omega}{c^2} E_x \\ -ik_x B_z = -\dfrac{i\omega}{c^2} E_y \\ ik_x B_y = -\dfrac{i\omega}{c^2} E_z \end{cases}$$

- Our wave equation is as usual

$$\vec{\nabla} \times \left(\frac{\vec{\nabla} \times \vec{E}}{i\omega} \right) = -\frac{i\omega}{c^2} \vec{E} \Rightarrow \nabla^2 \vec{E} = -\frac{\omega^2}{c^2} \vec{E}.$$

- For the x-component, $\dfrac{\partial^2 E_x}{\partial x^2} + \dfrac{\partial^2 E_x}{\partial y^2} + \dfrac{\partial^2 E_x}{\partial z^2} = -\dfrac{\omega^2}{c^2} E_x.$

$$\Rightarrow -k_x^2 E_x + \frac{\partial^2 E_x}{\partial y^2} + \frac{\partial^2 E_x}{\partial z^2} = -\frac{\omega^2}{c^2} E_x.$$

Try separation of variables. $E_x = A_0 e^{ikx} f(y) g(z)$.

$$\Rightarrow -k_x^2 E_x + e^{ikx} g(z) \frac{\partial^2 f(y)}{\partial y^2} + e^{ikx} f(y) \frac{\partial^2 g(z)}{\partial z^2} = -\frac{\omega^2}{c^2} E_x$$

$$\Rightarrow -k_x^2 + \frac{1}{f(y)} \frac{\partial^2 f(y)}{\partial y^2} + \frac{1}{g(z)} \frac{\partial^2 g(z)}{\partial z^2} = -\frac{\omega^2}{c^2}$$

The above equation can be true for arbitrary (y, z) points iff

$$\begin{cases} \dfrac{1}{f(y)} \dfrac{\partial^2 f(y)}{\partial y^2} = -\lambda_1^2 \\ \dfrac{1}{g(z)} \dfrac{\partial^2 g(z)}{\partial z^2} = -\lambda_2^2 \end{cases}$$

where λ_1, λ_2 are constants. We choose negative signs because positive choice will lead to $f(y), g(z)$ having exponentially decaying / increasing forms, which is inconsistent with boundary conditions. For example, if $\dfrac{1}{f(y)} \dfrac{\partial^2 f(y)}{\partial y^2} = +\lambda^2 \Rightarrow f(y) \propto e^{\pm \lambda y}$, can't vanish on both $y=0$ and $y=a$.

So $\begin{cases} \dfrac{\partial^2 f(y)}{\partial y^2} = -\lambda_1^2 f(y) \\ \cancel{\partial^2 f}\ \dfrac{\partial^2 g(z)}{\partial z^2} = -\lambda_2^2 g(z) \end{cases} \Rightarrow \begin{cases} f(y) = \sin \lambda_1 y \\ g(z) = \sin \lambda_2 z \end{cases}$ (We absorb the constants into A_0 in E_x.)

- Our wave equation simplifies to $k_x^2 + (\lambda_1^2 + \lambda_2^2) = \frac{\omega^2}{c^2}$.

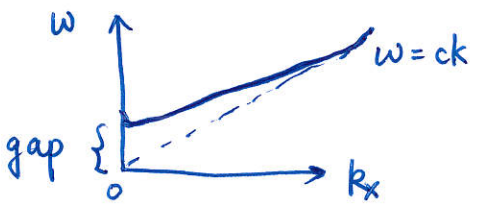
  If we further want to make $f(y)$, $g(z)$ satisfy boundary conditions,

  $$\begin{cases} f(a) = 0 \Rightarrow \lambda_1 = \frac{n_1 \pi}{a} \\ g(b) = 0 \Rightarrow \lambda_2 = \frac{n_2 \pi}{b} \end{cases} \quad \text{w/ } n_1, n_2 \in \mathbb{Z}^+.$$

  Thus $\frac{\omega^2}{c^2} = k_x^2 + \frac{\pi^2 n_1^2}{a^2} + \frac{\pi^2 n_2^2}{b^2}$, the frequency gets quantized.

- Take the minimum value $n_1 = n_2 = 1$, then if $\frac{\omega^2}{c^2} < \frac{\pi^2}{a^2} + \frac{\pi^2}{b^2}$, $k_x^2 < 0 \Rightarrow k_x$ is imaginary. This implies that the waveguide doesn't allow the propagation of transverse magnetic waves with small frequencies.

- The plot should be like:



At large k_x, the curve approaches the $\omega = ck$ line from above. Namely, for small wavelengths, our waves bounce up and down; they don't care about the walls. Across the waveguide there should be standing waves.

- Above we have only taken care of E_x. Will the corresponding E_y and E_z's also satisfy $\vec{E}_{\parallel} = 0$?

  So we have $E_x = A_0 e^{ik_x x} \sin\left(\frac{\pi n_1}{a} y\right) \sin\left(\frac{\pi n_2}{b} z\right)$. ~~Plug into~~ From The maxwell eqn $-ik_x B_z = -\frac{i\omega}{c^2} E_y \Rightarrow B_z = \frac{\omega}{c^2 k_x} E_y$. Plug into another eqn

  $$-ik_x E_y - \frac{\partial E_x}{\partial y} = i\omega B_z \Rightarrow -ik_x E_y - \frac{\partial E_x}{\partial y} = \frac{i\omega^2}{c^2 k_x} E_y$$

Substitute expression of E_x into the above equation,

$$-ik_x E_y - A_0 e^{ik_x x} \cos\left(\frac{\pi n_1}{a} y\right) \cdot \frac{\pi n_1}{a} \sin\left(\frac{\pi n_2}{b} z\right) = \frac{i\omega^2}{c^2 k_x} E_y$$

So $E_y \propto \cos\left(\frac{\pi n_1}{a} y\right) \sin\left(\frac{\pi n_2}{b} z\right)$.

$E_y(z=b) = 0$ is taken care of by $\sin\left(\frac{\pi n_2}{b} z\right)$.

$E_y(y=a) \neq 0$ is given by $\cos\left(\frac{\pi n_1}{a} y\right)$, but that is the orthogonal component; nonzero is fine.

Similar arguments for E_z. So everyone is consistent with boundary conditions.

- Next time we will further study the transverse electric wave case, probably. ~~We may~~ There is possibility that we will turn to the cavity resonators ~~which~~ where we confine our x-direction of the pipe to be finite as well.